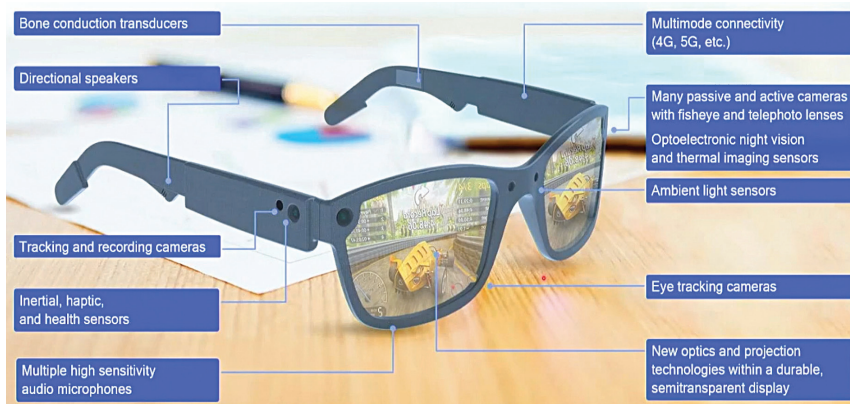




Features of XR glasses (Credit: Ken Wiseman, Qualcomm)

So, how do we keep pushing the form factor and what architecture do we use? These glasses necessarily have one chip, as we have in our handsets. According to Wiseman, we have to start distributing some chips throughout the XR glass. This way, we can make it not just thermally capable of processing what is needed but also ensure that the industrial design of the glass won't be compromised by thermal issues.

Smart cars. "The cockpits in newer cars are actually digital," explains Wiseman. A supersized version of the features present in mobile handsets is also present in smart cars. Although they have certain features that are not present in mobile phones—like multiple displays and ADAS—the core technology is similar. AI accelerators that are there in our handsets are again scaled with respect to the target market—automobiles.

"We can't just copy the architecture of mobiles and move it into the automobile space. There are other pieces of IP that are specific to some of these different applications," he adds. However, he believes that mobile architecture can be extended to add the features that we need for these target markets.

PCs, wearables, and consumer electronics. It is all about scaling. Some PC applications, like computing, are much superior to mobile ones, but some applications, like

Some important components of Smart Car Infrastructure

- AI
- Heterogeneous Computing
- Computer Vision
- Precise Positioning
- Audio
- Sensor Fusion
- Multi-node 4G/5G

cameras, may not be. It is possible to converge mobile architecture into these devices—after scaling, or even directly.

"I have a drone that has a Qualcomm chipset with Wi-Fi, and it has all the capabilities that are needed to enable multiple cameras," says Wiseman. Even exercise machines like Pelotons have tablet displays. The technology is not far off from what we have in a mobile phone today.

What next?

"The handset today has gotten limited in terms of volume and thermal capability. Moore's Law is slowing down, but we're going to keep milking it for as long as we can, but it's definitely becoming a challenge," says Wiseman. "So, there is going to be some transformation very soon!" **EFY**

This article has been transcribed and curated by EFY's Aaryaa Padhyegurjar from Ken Wiseman's keynote address at the VLSI Design Conference 2022.

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